Proof Workbook

Exercises, Guided Walkthroughs, and Thinking Prompts

Engineering Math to Pure Math

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1 How to Work These Problems

Do not only hunt the answer. The goal is to train your eye. For each exercise, write the claim, identify the definitions, choose a strategy, and leave a readable proof.

Focus

A good beginner proof often feels almost too slow. That is fine. Slow proofs become fast understanding later.

2 Set 1: Logic and Translation

Exercise 1. Translate each sentence into symbols. Then write its negation in symbols and in plain English.

1. Every integer has a square that is nonnegative.
2. Some real number is larger than its square.
3. For every programmer there is a bug that they cannot reproduce.
4. Every continuous function on $[0, 1]$ is bounded.
5. There is a prime number larger than 1000.

Exercise 2. Determine whether each statement is true or false. If false, give a counterexample.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x + y = 0$.
2. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}$ such that $x + y = 0$.
3. $\forall n \in \mathbb{Z},\ n^2 + n$ is even.
4. $\forall a, b \in \mathbb{Z},\ a \mid b \Rightarrow b \mid a$.

3 Set 2: Direct Proof

Exercise 3. Prove that if $a, b, c \in \mathbb{Z}$, $a \mid b$, and $b \mid c$, then $a \mid c$.

Walkthrough. Start from definitions. $a \mid b$ means $b = ak$ for some $k \in \mathbb{Z}$. $b \mid c$ means $c = b\ell$ for some $\ell \in \mathbb{Z}$. Substitute: $c = (ak)\ell = a(k\ell)$. Since $k\ell \in \mathbb{Z}$, $a \mid c$.

Exercise 4. Prove that the sum of two even integers is even.

Exercise 5. Prove that if $m, n \in \mathbb{Z}$ are odd, then mn is odd.

Exercise 6. Prove that if $x \in \mathbb{R}$ and $0 < x < 1$, then $x^2 < x$.

4 Set 3: Contrapositive and Contradiction

Exercise 7. Prove by contrapositive: if n^2 is even, then n is even.

Exercise 8. Prove that $\sqrt{2}$ is irrational.

Exercise 9. Prove that there are infinitely many prime numbers.

Walking question

For Euclid's proof of infinitely many primes, imagine the known primes as a fenced garden. What does multiplying them all and adding 1 do to the fence?

5 Set 4: Sets

Exercise 10. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Proof skeleton.

1. To prove equality, prove both subset directions.
2. For $\subseteq$, let $x \in A \cap (B \cup C)$.
3. Then $x \in A$ and $x \in B \cup C$.
4. Split into cases: $x \in B$ or $x \in C$.
5. Finish the first direction, then reverse the reasoning for $\supseteq$.

Exercise 11. Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Exercise 12. Give an example where $A \subsetneq B$ and $\mathcal{P}(A) \subsetneq \mathcal{P}(B)$.

6 Set 5: Functions

Exercise 13. Let $f : A \to B$ and $g : B \to C$. Prove that if f and g are injective, then $g \circ f$ is injective.

Exercise 14. Let $f : A \to B$ and $S, T \subseteq B$. Prove

$$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

Exercise 15. Find a function $f : \mathbb{R} \to \mathbb{R}$ and sets $S, T \subseteq \mathbb{R}$ such that

$$f(S \cap T) \neq f(S) \cap f(T).$$

7 Set 6: Relations

Exercise 16. Define $a \sim b$ on $\mathbb{Z}$ by $a \sim b$ if $a - b$ is divisible by 5. Prove $\sim$ is an equivalence relation.

Exercise 17. Let R be a relation on a set A. Write the definitions of reflexive, symmetric, antisymmetric, and transitive. Give one relation that has each property and one that fails each property.

8 Set 7: Induction

Exercise 18. Prove that for every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Exercise 19. Prove that for every $n \geq 1$,

$$2^n \geq n + 1.$$

Exercise 20. Use strong induction to prove every integer $n \geq 2$ is a product of primes.

9 Set 8: Countability

Exercise 21. Prove that the set of even integers is countably infinite.

Exercise 22. Prove that $\mathbb{Z}$ is countable.

Exercise 23. Prove that $\mathbb{Q}$ is countable. Hint: arrange rational numbers by numerator-denominator grid and skip repeats.

Exercise 24. Explain Cantor's diagonal argument in your own words. Do not use the phrase "it is obvious."

10 Set 9: First Analysis Problems

Exercise 25. Prove directly from the definition that $\lim_{n\to\infty} 1/n = 0$.

Exercise 26. Prove that if $a_n \to a$ and $b_n \to b$, then $a_n + b_n \to a + b$.

Exercise 27. Prove that $f(x) = 3x + 2$ is continuous at every $c \in \mathbb{R}$ using the ε-δ definition.

11 Set 10: First Algebra Problems

Exercise 28. Prove that the identity element of a group is unique.

Exercise 29. Prove that every element of a group has a unique inverse.

Exercise 30. Let G be a group and $a, b \in G$. Prove $(ab)^{-1} = b^{-1}a^{-1}$.

12 Portfolio Assignment

Choose twelve exercises from this workbook: two logic, two set/function, two relation/induction, two countability/analysis, two algebra, and two wild cards. Rewrite them as a polished proof portfolio. For each proof, include:

- the claim,

- the definitions used,
- the proof strategy,
- the final proof,
- one sentence about the main idea.